DHCP Lab

TITLE : DHCP Configuration in Cisco Packet Tracer

Aim : To configure a router as a DHCP server so that PCs can automatically obtain IP addresses.

NETWORK DEVICES USED :

1. Router – 1

2. Switch – 1

3. PC – 4 (PC0, PC1, PC2, PC3)

4. Data Cables – Copper Straight Through

NETWORK ADDRESSING :

- Network Address : 192.168.10.0
- Subnet Mask : 255.255.255.0
- Router Interface (Connected to Switch) : 192.168.10.1

PROCEDURE :

Step 1 : Create Network Topology

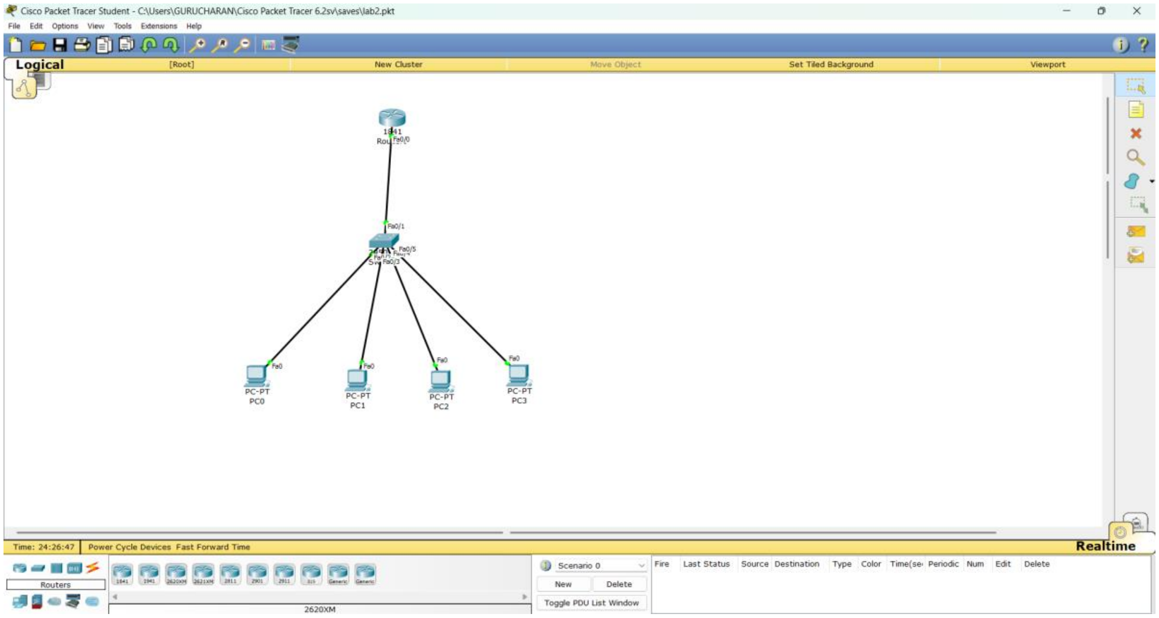
1. Open Cisco Packet Tracer.

2. Add 1 Router, 1 Switch and 4 PCs.

3. Connect all devices using Copper Straight Through cables.

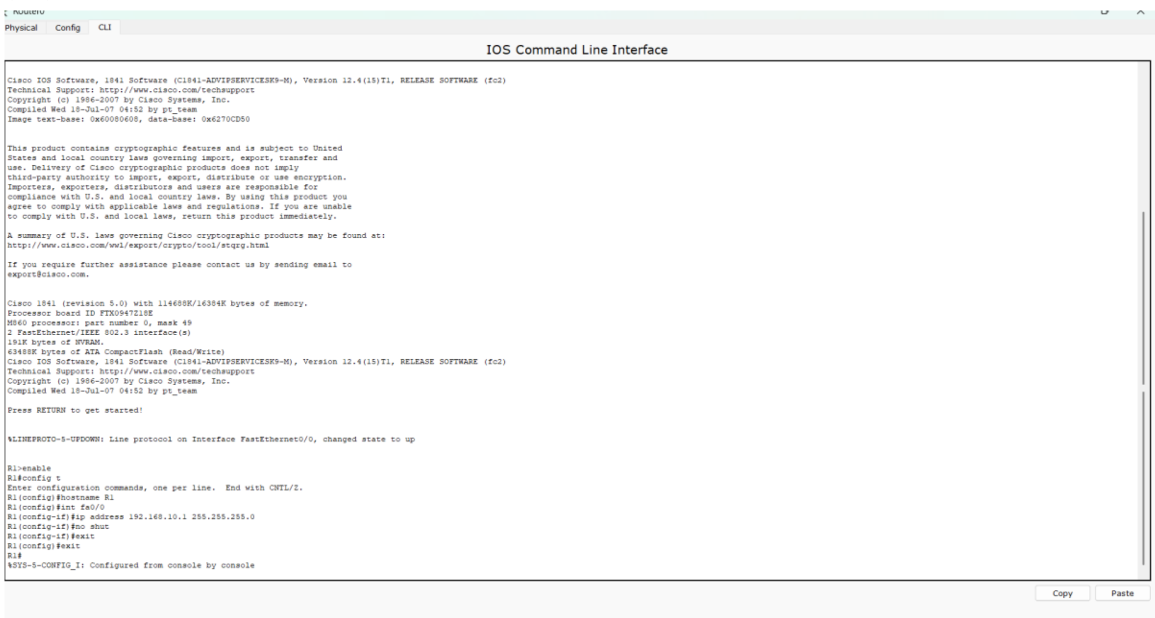
Connections :

- PC0 → Switch
- PC1 → Switch
- PC2 → Switch
- PC3 → Switch
- Router → Switch



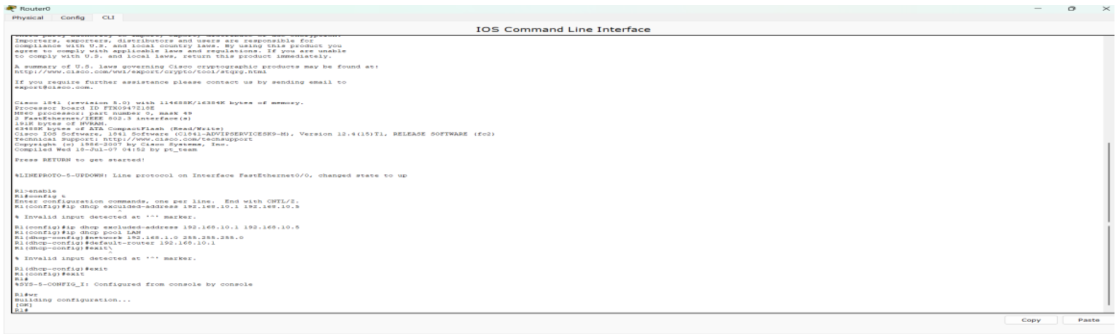
Step 2 : Configure Router IP Address

- Open Router CLI and enter the following commands :
- enable
- configure terminal
- interface fa0/0
- ip address 192.168.10.1 255.255.255.0
- no shutdown
- exit



Step 3 : Configure Router as DHCP Server

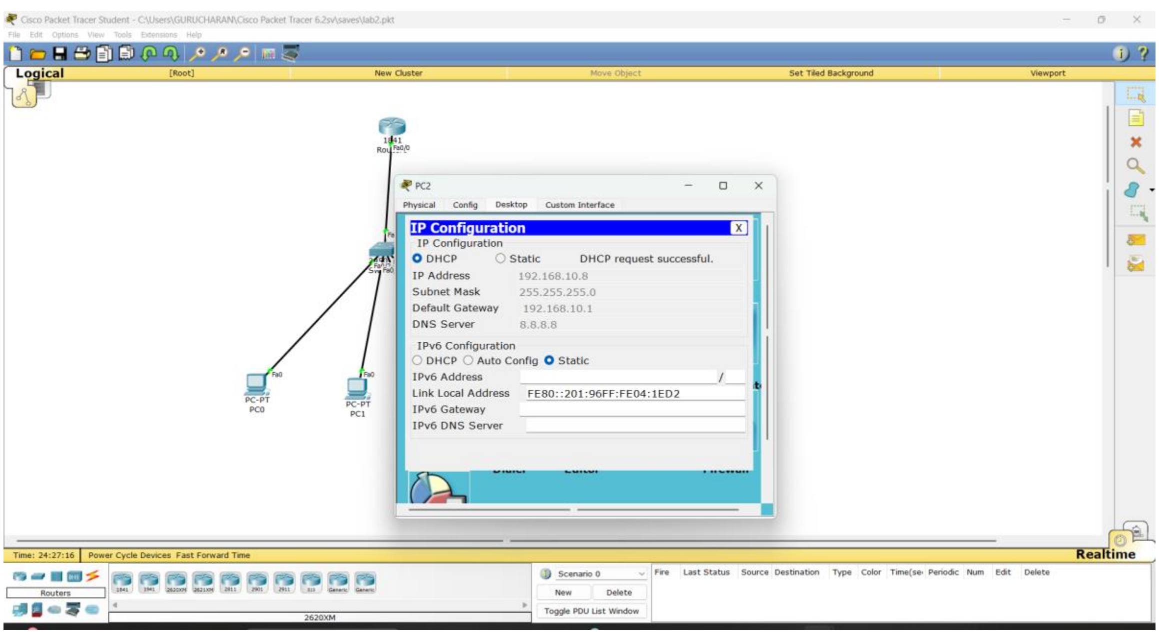
- configure terminal
- ip dhcp excluded-address 192.168.10.1 192.168.10.5
- ip dhcp pool LAN
- network 192.168.10.0 255.255.255.0
- default-router 192.168.10.1
- exit



Step 4 : Configure PCs to Obtain IP Automatically Open each PC (PC0, PC1, PC2, PC3).

- Go to Desktop → IP Configuration.
- Select DHCP option.

The PC will automatically receive an IP address from the router.



Step 5 : Verification

- Check IP address assigned to each PC.
- Expected range : 192.168.10.6 – 192.168.10.254.

Step 6 : Ping Test

- Open Command Prompt on any PC.
- Type ping followed by another PC IP address.

Example : ping 192.168.10.6

If replies are received, the network configuration is successful.

EXPECTED RESULT :

All PCs should automatically receive IP addresses from the DHCP server in the range 92.168.10.6 – 192.168.10.254 and should be able to communicate with each other.